



LINEAR ALGEBRA

UE19MA251

APARNA B S

Department of Science and Humanities

LINEAR ALGEBRA – UE19MA251

Agenda – Problems on SVD



■ Example 2

LINEAR ALGEBRA – UE19MA251

Agenda – Problems on SVD



Example 2. Find the SVD for $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & 2 \end{bmatrix}$

Soln: SVD: $A = U \Sigma V^T$

$$AA^T = \begin{bmatrix} 9 & -1 \\ -1 & 9 \end{bmatrix}$$

LINEAR ALGEBRA – UE19MA251

Agenda – Problems on SVD



Eigenvalues of AA^T are

$\lambda_1 = 10, 8$ (arranged in increasing order)

Corresponding Eigenvectors are

$$x_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad \text{and} \quad x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

LINEAR ALGEBRA – UE19MA251

Agenda – Problems on SVD

Clearly x_1 and x_2 are orthogonal.

$$V = \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = V^T$$

$$\Sigma = \begin{bmatrix} \sqrt{10} & 0 \\ 0 & \sqrt{8} \\ 0 & 0 \end{bmatrix}$$

LINEAR ALGEBRA – UE19MA251

Agenda – Problems on SVD



$$u_i = \frac{Av_i}{\sigma_i} ; i = 1, 2, 3 ; \sigma_1 = \sqrt{10} , \sigma_2 = \sqrt{8}$$

$$u_1 = \begin{bmatrix} -1/\sqrt{5} \\ 2/\sqrt{5} \\ 0 \end{bmatrix} \quad u_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$u_1^T u_2 = 0 ; u_1 \perp u_2 .$$

LINEAR ALGEBRA – UE19MA251

Agenda – Problems on SVD

using orthogonality condition, u_3 can be obtained. Let $u_3 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$\begin{bmatrix} -1/\sqrt{5} & 2/\sqrt{5} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

LINEAR ALGEBRA – UE19MA251

Agenda – Problems on SVD



$\therefore u' = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$, normalizing, we get

$$u_3 = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \\ 0 \end{bmatrix} ; u = \begin{bmatrix} -1/\sqrt{5} & 0 & 2/\sqrt{5} \\ 2/\sqrt{5} & 0 & 1/\sqrt{5} \\ 0 & 1 & 0 \end{bmatrix}$$

$$\therefore A = u \Sigma v^T$$



THANK YOU

Aparna B. S

Department of Science & Humanities

aparnabs@pes.edu